

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSE-AI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : 9

Team Members :

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1. Problem Title : **Employee Payroll Management System**

2. Problem Description : Managing payroll manually takes a lot of time and is prone to mistakes. The proposed system automates salary calculations, payroll processing, and report generation to improve efficiency and reduce errors.

3. Identified Entities (Classes)

Class Name	Description
Person	Stores basic info like id and name
Employee	Stores employee details and salary info
HR	Manages employee data and payroll
Manager	Approves salary payments
PayrollSystem	Handles salary calculation and processing
SalarySlip	Generates salary reports

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(baviours)
Person Class	id, name	displayBasicInfo(), displayRole()
Employee Class	empId, name, designation, basicSalary	viewSalary(), updateDetails()
HR Class	hrId, name	addEmp(), updateEmp(), processPayroll()
Manager Class	managerId, name	approveSalary()
PayrollSystem	payrollId, employeeList	calculateSalary(), generatePayslip()
SalarySlip Class	empId, salary, deductions, netSalary	generateReport()